

State Management & Resource Optimization

1. Hybrid State Persistence

SankeyLogic separates **structural metadata** from the **analytical engine state**. The Excel workbook only stores the "blueprint" of your model, while the high-performance graph objects exist exclusively in-memory during the active session.

- **How it works:** We persist only essential mapping parameters (UIDs, range pointers, and timestamps).
- **The Workflow:** Upon reopening a workbook, the chart is initially in a "dormant" state. Clicking **Update** re-initializes the engine, performing a full graph traversal and rebuilding the visualization from the latest source data.
- **Why this is optimal:** It prevents workbook bloat and eliminates the risk of "ghost data" (visuals being out of sync with cell values). It also ensures that complex analytical relationships are not permanently hardcoded into the file, keeping the .xlsx file clean and portable.

2. Single-Active-Analysis Model

To maintain the sub-second latency required for **Real-Time Downstream Propagation** and **Flow Integrity** calculations, the engine focuses its computational resources on a single active model.

- **Multi-Sheet Handling:** The analytical engine tracks the model on your current worksheet.
- **Best Practice:** When switching between worksheets containing different Sankey models, a quick **Refresh** is required to re-align the in-memory engine with the specific parameters of the new sheet. This ensures that terminal shares and peer-to-peer metrics are 100% accurate for the visible data.

3. Engine Dependency & Portability

The interactive analytical features (hover tooltips, tracing, downstream metrics) require the SankeyLogic Add-in to be active.

- **Distribution:** Since the engine is session-based, sharing a workbook with users who don't have the Add-in will result in a static reference.
- **Snapshots:** For reporting, documentation, or sharing with non-users, use the **Export to Image** function. This creates a high-fidelity visual record of the analysis while maintaining the security and integrity of the underlying live model.